LabaLab
Language, Affect and Behaviour in AI

STUDENT GUIDE · NOVEMBER 2026

Lab robots, step by step

How to switch on, connect and run every robot in room −132.



LabaLab · BGU

THE LAB

Five kinds of robot, one app



Reachy Mini

Several units, named on the robot: reachy1, reachy2 …

Connects by **WiFi**

Main button **Talk**



Reachy Mini Lite

The small one with no WiFi of its own.

Connects by **USB cable**

Main button **Connect**



Furhat

A face projected onto a mask.

Connects by **WiFi**

Main button **Open Studio**



Pepper

Tall, on wheels, with a tablet on its chest.

Connects by **WiFi**

Main button **Open dashboard**



NAO

Small humanoid. Runs the lab's NAO app.

Connects by **WiFi**

Main button **NAO app (staff)**

The **LabaLab Robot Hub** app finds every robot on the WiFi, connects it and opens the right program for it.

BEFORE ANYTHING ELSE

Three rules that prevent most problems

1

WiFi first, then the robot

Robots look for WiFi while they start up. Furhat never looks again, so a late WiFi means a deaf, silent Furhat.

2

Laptop on the robots' WiFi

In the lab that is **HOT-WiFi** (open, no password). Turn off VPN, and auto-join for any other network.

3

Charge before every session

Reachy can't report its battery: it just stops mid-sentence. Pepper must never run flat, or it can lock for good.

ONCE PER LAPTOP

Install Robot Hub

1. **Open PowerShell**
   Press Start, type PowerShell, press Enter.

2. **Paste the line below, press Enter**
   Exactly as written, in one piece.

3. **Wait for the app to open**
   About 114 MB. A LabaLab Robot Hub icon lands on the desktop.

4. **Add a key**
   Paste it in **Your key**, press **Test and save**, wait for **It works**.

Good to know

Windows 10 or 11. **No admin rights**, Python or Git needed.

Line blocked? Download the zip from the latest GitHub release, unzip it, double-click **Install Robot Hub.cmd**.

Keys: each student gets their own capped key from lab staff.

```
irm https://github.com/labbgulab-lab/robot-hub/releases/latest/download/install.ps1 | iex
```

On GitHub: **github.com/labbgulab-lab/robot-hub** (Releases has the zip).

EVERY SESSION

The same six steps, in this order

1 Charge the robot

Full battery before you start. Reachy and Pepper can’t be trusted to warn you.

2 Laptop on HOT-WiFi

Or your phone hotspot for a demo outside the lab. VPN off.

3 Switch the robot on

Wait until it is fully up. Furhat takes several minutes.

4 Open Robot Hub

Double-click **LabaLab Robot Hub** on the desktop.

5 Wait, then Connect

About 30 s. Each robot shows as a card marked **Available**. Press **Connect**.

6 Press its main button

Talk (Reachy), **Open Studio** (Furhat), **Open dashboard** (Pepper).

To stop everything: **Settings › About › Quit**

REACHY MINI

Talk to one Reachy

1. **Charge it fully, then switch it on**
   It has no battery reading at all.
2. **Connect it in Robot Hub**
   Its card shows its name, e.g. reachy3.
3. **Check Speaks through on the card**
   It must be a Gemini or GPT key.
4. **Press Talk**
   First time only: it installs its app, a few minutes.
5. **Press Start conversation**
   The robot speaks first.
6. **To end, say “goodbye”**
   Or press **End conversation**.

Install once

Pollen’s **Reachy Mini app**, on every Reachy student’s laptop:
hf.co/reachy-mini/#/download

Watch out

It hears with its own mic. Speak close to it. Group talk can use the DJI Mic Mini receiver in the laptop.

Wizard of Oz (type a line, press **Say**) needs a Gemini key on the robot.

REACHY MINI

Group conversation

1. **Connect two or more Reachys**
2. **Press Group talk**
   In the hub’s Reachy app band. The Group tab opens.
3. **Tick Included for each robot**
4. **Write a Topic, pick Language and Minutes**
5. **Press Get ready, then a mode**
   Group talk: robots debate, people join in. **Interrupt**: a person talks, robots cut in. **Free talk**: no topic.
6. **Press Stop to end**
   They say a closing line. **Stop now** ends it at once.

Tips

Steer: type in the Steer box and press Enter to nudge the next line.

Choir: all robots speak as one voice, from the next start.

Disconnect first before you change the robots’ order, Included or Listens.

Transcripts: Settings › Conversations › Open folder.

REACHY MINI LITE

The USB Reachy

1. **Install the Reachy Mini app, once**
   From **hf.co/reachy-mini/#/download**, or press **Set up** on the Lite's card.
2. **Plug in power, then USB**
   Its power supply into the wall, its USB cable into the laptop.
3. **Keep the Reachy Mini app running**
   It is the Lite's brain. Unplug the Lite and the app closes about 30 s later.
4. **Connect, then Open › Play on Conversation**
   To stop, say "go to sleep".

⚠ Watch out

No WiFi at all. It never shows up on a network scan, only over USB.

Windows only in Robot Hub.

The desktop app holds one robot. Keep it on the Lite. A wireless Reachy can run at the same time: Robot Hub talks to that one directly.

FURHAT

Start Furhat, the order matters

1. **WiFi on first**
   Furhat looks for WiFi only while it starts. Hotspot: 2.4 GHz.
2. **Mic plugged in before power**
   It only finds the microphone while it starts.
3. **Switch on and wait**
   Several minutes. The face on the mask means it is ready.
4. **Connect, then Open Studio**
   Studio password: in the lab’s Bitwarden.
5. **Reset the voice engines, every time**
   Settings › Speaking, then Listening: switch each engine to Custom and straight back to Furhat provided.
6. **Check language and hearing**
   English (United States). Recognition: Microsoft Azure.

If it goes wrong

Lips move, no sound
The engines didn’t load. Do step 5.

It can’t hear you
Recognition must be Microsoft Azure, never Google.

English in a Japanese voice
Factory voice is Japanese. Set the language.

Fine yesterday, silent today
It started before the WiFi. WiFi on, restart Furhat.

PEPPER · 1 OF 2

Switch Pepper on and off safely

1. **Clear 90 cm around it**
   It moves as it wakes up.

2. **Press the chest button once**
   About 2 minutes. Ready when its eyes are white.

3. **Press it once more**
   It says its name and IP. “No IP” means no network.

4. **To turn off: hold the chest button 4 s**
   Only when it stands in a safe spot, or it may fall.

⚠ Safety first

29.6 kg, on wheels, and it can fall forward.

Never let the battery run flat: it can lock for good. Charge it for hours.

Walking only with the charger unplugged and the charging flap closed. Keep about 1.5 m clear.

Never hold 4 s while it is off: that starts a factory mode.

PEPPER · 2 OF 2

Run the Pepper demo

1. **Set up once**
   Card says **Needs setup**: Set up › Get it (about 360 MB) › Done.
2. **Pick its keys on the card**
   The demo needs a GPT key and an ElevenLabs key.
3. **Connect, then Open dashboard**
   It’s for you. Visitors only see Pepper’s chest tablet.
4. **Pick Feelings Together (demo)**
   The scripted demo: 9 steps, Hebrew or English.
5. **Press Start conversation**
   Pepper speaks first, then waits for a tap or an answer.
6. **Press Stop everything when done**
   Top right of the dashboard.

Good to know

Wizard of Oz: Say a quick phrase or type one. Move: wave, nod, arms, a short walk.

Practise with no robot: More › Rehearse.

Day before a demo: More › Tablet emoji check.

Tablet went dark? Press its own power button, on its edge.

Never use the tablet’s Advanced Settings or remove the vendor’s ht_ apps.

NAO

Start NAO and its app

1. **Press the chest button once**
   NAO turns on. Press again and it says its address.
2. **Set up once in Robot Hub**
   The same 360 MB kit as Pepper: Set up › Get it.
3. **Connect NAO on its card**
   Then **Sit**, **Lie down** or **Stand up** (press Confirm, someone beside it).
4. **Press NAO app (lab staff)**
   Needs staff tools on and the lab's NAO folder.
5. **At the end: press Lie down**
   Then hold the chest button 3 s. Never the crouch posture: NAO fell.

⚠ Watch out

Microphone clash. The NAO app listens through Windows' default mic. If a Reachy holds it, Robot Hub refuses and says who has it.

First try fails? Normal. NAO answers before it is fully ready. Try again.

Without the hub: run deploy_nao.py once per boot, then main.py, and open localhost:8000.

ONCE PER NETWORK

Teach a robot a new WiFi

Robot	How
Reachy Mini	Robot Hub, staff tools on: the robot's **…** › **Add WiFi** › Join. A brand-new robot: lab staff run tools\reachy_bringup.py.
Pepper	Robot Hub, staff tools on: Pepper's **…** › **Add WiFi**. This teaches the head only: lab staff set the tablet separately.
NAO	Robot Hub, staff tools on: NAO's **…** › **Add WiFi**. If it knows no WiFi: a cable, then lab staff use connmanctl.
Furhat	Furhat Studio › Settings › Network › **Add Network**, then restart. Knows no WiFi: keyboard and mouse on its own screen.
Reachy Mini Lite	Nothing to do. It is USB only.

Robots remember a WiFi by **name and password**. Use the same hotspot name and password on every lab phone, and every robot knows them all.

OUTSIDE THE LAB

A demo on a phone hotspot

1. **Teach every robot the hotspot, in the lab**
   Add WiFi in each robot’s details (staff tools).
2. **Make the hotspot 2.4 GHz**
   iPhone: Maximize Compatibility on. Android: band 2.4 GHz.
3. **Give it a plain name**
   Letters, digits and spaces only. An iPhone’s “Name’s iPhone” breaks Reachy.
4. **Count devices: 5 at most**
   Laptop 1, Pepper 2 (head and tablet), every other robot 1.
5. **Hotspot on before any robot**
   Keep the hotspot screen open while robots join.
6. **Laptop on the hotspot, open Robot Hub**
   A “not HOT-WiFi” note is expected here.

Robot Hub plans it for you: **Settings › Demo on a phone hotspot › Plan**.

WHEN SOMETHING GOES WRONG

What you see, and what to do

What you see	What to do
No robots nearby	Is the laptop on the robots' WiFi? Press Open WiFi settings. Turn off auto-join for other WiFi.
Every robot vanished at once	The laptop slipped onto another WiFi, often campus. Rejoin the robots' WiFi.
VPN on	Turn the VPN off. Robots may not show up with it on.
Card says Not nearby	Switch the robot off and on, then press Scan.
Card says In use	Another laptop has it. Ask that person, or lab staff press Take over.
Reachy moves, never replies	Does Speaks through have a working key? Speak close to the robot: it hears with its own mic.
No key yet, or Used up	Press Add a key. Used up means no credit left: ask lab staff.
Couldn't start	Press See log, then Try again on the card.
Nothing helps	Logs › Save a report for the lab. It goes to Downloads with keys removed. Send it.

KEEP THIS SLIDE

Where everything lives

Robot Hub app and guides	`github.com/labbgulab-lab/robot-hub`
Student and staff guides	`robot-hub/docs/guides`
Reachy app	`labbgulab-lab/reachy-mini-conversation-app-bgu-lab`
Reachy Mini app (Pollen)	`hf.co/reachy-mini/#/download`
NAO app	`labbgulab-lab/antagonistic-robot`
Each robot, measured	`robot-hub/profiles`
Passwords	The lab’s Bitwarden vault

A student stuck? Ask for **Logs › Save a report for the lab**. It has everything, with keys removed.

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